INTERNOVA INTERNSHIP

WEEK 5 PROJECT REPORT

Power BI & Business Intelligence

Student	Tiwari Aman Dhananjay
Project	Sales Performance & Business Intelligence Dashboard
Tool	Microsoft Power BI Desktop
Core Skills	Power Query • Data Modeling • DAX • Data Visualization
Submission	Week 5 – Power BI & Business Intelligence

A practical Power BI project covering data import, data preparation, relational modeling, DAX measures, interactive visualizations and business recommendations.

1. Project Overview

This Week 5 project demonstrates a complete Business Intelligence workflow in Microsoft Power BI. Structured sales, customer and product data are transformed into an interactive management dashboard. The work covers practical stages expected from an entry-level data analyst: preparing data, connecting tables, creating calculations, selecting suitable visuals and converting results into business observations.

Project objective

• Import and inspect structured business datasets in Power BI.

• Prepare sales data using Power Query concepts.

• Build relationships between Customers, Products and Sales.

• Create reusable DAX measures for business KPIs.

• Design an interactive one-page dashboard.

• Interpret results and provide practical recommendations.

Dashboard KPI snapshot

Metric	Observed result
Total Sales	Approximately 9M
Total Records	118
Average Order Value	78,828.86
Highest-performing region	East
Highest-performing category	Computers

Note: Values are reported as displayed in the completed Power BI dashboard. Total sales is displayed in a compact rounded format (9M).

2. Data Sources & Dataset Structure

The project uses three logical business tables. Sales_Clean acts as the central transaction table, while Customers and Products provide descriptive attributes used for slicing and analysis.

Datasets used

Table	Purpose	Important fields
Customers	Customer master data	Customer_ID, Customer_Name, Region, Segment
Products	Product master data	Product_ID, Product, Category, Unit_Price
Sales_Clean	Sales transactions	Sale_ID, Date, Customer_ID, Product_ID, Quantity, Discount, Sales_Amo

Data preparation approach

• Sales data was prepared in a clean tabular structure for numeric, date and categorical analysis.

• Customer and product master tables were retained separately to support one-to-many relationships.

• The final model uses Sales_Clean as the central transaction table.

• Region and Category are used as business dimensions for interactive analysis.

Relationship model

Customers (1) ■■■■■ < (many) **Sales_Clean** (many) > ■■■■■ (1) **Products**

3. Task-wise Implementation

The assignment was completed through seven practical stages. Each stage contributes to the final dashboard and demonstrates a different Power BI capability.

Task 1 – Power BI Introduction & Data Import

• Power BI Desktop was used as the main BI environment.

• Customer, product and sales data were loaded into the report.

• The imported fields were reviewed before modeling.

Task 2 – Data Cleaning & Preparation using Power Query

• The sales dataset was inspected as the main transaction source.

• Column structure and data types were checked before analysis.

• The clean sales table was used for the final dashboard.

Task 3 – Data Modeling & Relationships

• Customers[Customer_ID] is related to Sales_Clean[Customer_ID].

• Products[Product_ID] is related to Sales_Clean[Product_ID].

• The model follows a one-to-many structure from dimension tables to sales.

Task 4 – DAX Calculated Measures

• Measures were created for sales, averages, record count, quantity and transaction extremes.

• A percentage-of-total measure was also prepared for analytical use.

Task 5 – Power BI Visualizations

• KPI cards show high-level performance.

• Bar, line and donut charts provide category, time and region analysis.

• A product table provides detailed values.

Task 6 – Interactive Dashboard & Filters

• Region and Category slicers allow interactive filtering.

• The report combines summary and detail views on one page.

Task 7 – Business Insights & Recommendations

• Dashboard results were interpreted from a management perspective.

• The strongest region/category were identified and actions were proposed.

4. DAX Measures & KPI Logic

DAX measures make the dashboard dynamic. The measures respond when the user changes a slicer or interacts with a visual.

```
Total Sales = SUM(Sales_Clean[Sales_Amount])
Average Sales = AVERAGE(Sales_Clean[Sales_Amount])
Total Records = COUNTROWS(Sales_Clean)
Maximum Sale = MAX(Sales_Clean[Sales_Amount])
Minimum Sale = MIN(Sales_Clean[Sales_Amount])
Total Quantity = SUM(Sales_Clean[Quantity])
Average Order Value = DIVIDE([Total Sales], [Total Records])
Sales % of Total = DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(Sales_Clean)))
```

How the measures support the dashboard

- **Total Sales:** primary business KPI for overall sales performance.
- **Total Records:** shows the number of transactions included.
- **Average Order Value:** provides an average transaction benchmark.
- **Total Quantity:** shows product volume.
- **Maximum/Minimum Sale:** shows the transaction-value range.
- **Sales % of Total:** supports contribution/share analysis.

Current KPI values

KPI	Dashboard value	Purpose
Total Sales	≈ 9M	Overall sales performance
Total Records	118	Transaction count
Average Order Value	78,828.86	Average sales per record

5. Dashboard Design & Visual Analysis

The final dashboard follows a simple analytical flow: KPIs first, category and product analysis next, followed by time and regional analysis, with slicers available for interactive filtering.

KPI Cards

• Total Sales card provides the overall sales figure.

• Total Records card displays 118 transaction records.

• Average Order Value card displays 78,828.86.

Sales by Category

The bar chart compares sales across product categories. Computers is the strongest category in the completed dashboard, making it an important area for inventory and marketing attention.

Sales by Month

The line chart provides a month-wise view of sales performance. It can support planning, stock allocation and campaign timing.

Sales by Region

The donut chart shows regional contribution. East is the largest contributor in the completed dashboard, at approximately 58.9% of the displayed sales total.

Product Detail Table

The product table provides product-level quantity and total-sales values, adding detail to the summary charts.

Interactive Filters

• **Region:** filters the report by geographic market.

• **Category:** filters the report by product category.

These slicers make the report interactive rather than static and allow focused analysis without rebuilding the report.

6. Business Insights

The dashboard converts the prepared data into observations that can be understood without reading the underlying tables.

No.	Insight	Business meaning
1	Overall sales are approximately 9M.	The dashboard shows substantial sales volume across the analyzed records.
2	118 sales records are included.	The KPI provides a clear measure of transaction volume.
3	Average Order Value is 78,828.86.	This can be used as a benchmark for future performance.
4	East is the largest regional contributor.	The region is a major market and deserves continued attention.
5	Computers is the leading category.	Inventory and marketing can prioritize this category.

7. Business Recommendations

Recommendation 1 – Protect the strongest market

The East region contributes the largest share of dashboard sales. Management can maintain product availability, monitor demand and continue effective campaigns in this region.

Recommendation 2 – Prioritize the strongest category

Computers is the leading category. Inventory planning, promotions and product visibility can prioritize this category to protect performance.

Recommendation 3 – Investigate weaker segments

Lower-performing regions and categories can be reviewed using the slicers. Targeted promotions, bundles or customer-specific offers can then be tested.

8. Skills Demonstrated & Project Workflow

This project demonstrates a practical Business Intelligence workflow and connects technical Power BI skills with business interpretation.

Skill area	Application in project
Power BI Desktop	Report creation, visual selection and dashboard layout
Power Query	Data inspection and preparation
Data Modeling	Relationships between customer, product and sales tables
DAX	Measures and KPI calculations
Data Visualization	Bar, line, donut, cards and table visuals
Interactive Reporting	Region and Category slicers
Business Analysis	Insights and recommendations

End-to-end workflow

Source Data → Import → Power Query Preparation → Data Model → Relationships → DAX Measures → Visualizations → Slicers → Business Insights → Recommendations

Project deliverables

- Power BI report (.pbix).
- Datasets used for the project.
- Power BI support/reference files.
- Detailed project report.
- Final dashboard screenshot for evidence.

9. Conclusion & Submission Checklist

Conclusion

The Week 5 Power BI project demonstrates how business data can be transformed into a usable Business Intelligence dashboard. The completed report combines data preparation, relational modeling, DAX calculations, visual analysis and interactive filtering in a single reporting page.

The final dashboard provides both a management summary and detailed analytical views. KPI cards, category and regional charts, monthly trends, product details and slicers make the report suitable for demonstrating practical Power BI skills in an internship submission.

Final submission checklist

✓	Power BI dashboard completed
✓	KPI cards added
✓	Category bar chart added
✓	Monthly line chart added
✓	Regional donut chart added
✓	Product detail table added
✓	Region and Category slicers added
✓	DAX measures created
✓	Relationships created
✓	PBIX file saved
✓	Final dashboard screenshot prepared
✓	Detailed PDF report prepared

Prepared by

Tiwari Aman Dhananjay
Internova Internship – Week 5
Power BI & Business Intelligence

End of Report